Silicon Photonics & Integrated Optics

Module 2.1: Fundamentals of All-Pass Silicon Microring Resonators

Module Summary

This module introduces the fundamental physics and mathematical framework governing **silicon microring resonators** operating in the all-pass (notch filter) configuration. Benefiting from high refractive index contrast ($n_{Si} \approx 3.45$ vs. $n_{SiO_2} \approx 1.45$), silicon photonic waveguides can bend light tightly ($R < 5\,\mu$m) with minimal radiation loss. We derive from first principles the complex amplitude transfer function **Eq**. (**1**) and power transmission ratio **Eq**. (**2**), detailing the 90° quadrature coupling phase shift via unitary scattering matrices, infinite geometric series recirculation, and the critical coupling resonance condition.

1 Physical Intuition & Driving Analogies

1.1 The Highway and the Racetrack

Imagine a straight highway (**the bus waveguide**) running alongside a circular racetrack (**the microring resonator**). Light travelling down the bus waveguide naturally continues straight. However, when the ring is brought into sub-micron proximity to the bus waveguide, a portion of the optical field overlaps across the narrow dielectric gap and "leaks" into the ring via **evanescent coupling**.

Once trapped inside the ring, light recirculates around the loop perimeter and returns to the coupling junction:

- **Off-Resonance:** Light returning from the ring is out of phase with incoming bus light. Interference is incomplete, and most light exits the output port undisturbed.
- **On-Resonance:** Light completes a round trip in exact phase sync ($\phi = 2m\pi$). The light leaking back into the bus waveguide is 180° out of phase with the uncoupled bus field, leading to **destructive interference**. This produces a sharp drop or "notch" in transmitted optical power at resonant wavelengths.

1.2 The Driven Harmonic Oscillator Analogy

To understand the phase dynamics, consider the classic mechanical driven oscillator (such as a mass-spring system or pendulum):

1. **Resonant Phase Lag:** In classical mechanics, driving a harmonic oscillator precisely at its natural resonance frequency causes the displacement response of the mass to lag behind the driving force by 90° ($\pi/2$ rad).
2. **Coupled Wave Phase Shift:** In integrated optics under time-harmonic convention ($e^{j\omega t}$), the evanescent field in the bus waveguide acts as the driving force, while the ring field acts as the driven oscillator. The field coupled across the gap inherently acquires a 90° phase lag.
3. **Mathematical Representation:** In complex algebra, a 90° phase rotation corresponds to multiplication by the imaginary unit j (since $e^{j\pi/2} = j$).

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2 Key Concepts & Terminology

Microring Resonator A closed-loop dielectric waveguide in which light recirculates continuously, forming stationary optical modes at specific resonant frequencies.

Bus Waveguide A straight or gently curved optical waveguide positioned adjacent to the microring to inject input signals and collect transmitted signals.

Coupling Gap The nanometer-scale physical separation (typically 100–300 nm) between the bus waveguide and the microring core.

Evanescent Coupling Non-radiative energy transfer between neighboring waveguides via the exponentially decaying electromagnetic tail extending beyond the core boundaries.

Quadrature Phase Shift (j) The fundamental 90° ($\pi/2$ rad) phase shift acquired by an optical wave when cross-coupling into an adjacent waveguide.

All-Pass Configuration A ring resonator coupled to a single bus waveguide, serving as a notch filter (light is either transmitted straight through or extinguished at resonance).

Round-Trip Length (L) The total physical perimeter distance traveled by light in one complete turn ($L = 2\pi R$ for a circular ring of radius R).

Round-Trip Phase Shift (ϕ) The cumulative phase angle accumulated over one round trip: $\phi = \beta L = \frac{2\pi n_{\text{eff}}}{\lambda} L$.

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3 Rigorous Proof: Quadrature Phase Shift (j) via S-Matrix

To prove why cross-coupling requires a factor of j, consider the **Scattering Matrix (S-matrix)** of a symmetric, lossless 2×2 directional coupler:

$$\begin{pmatrix} E_{\text{out},1} \\ E_{\text{out},2} \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} E_{\text{in},1} \\ E_{\text{in},2} \end{pmatrix} = \begin{pmatrix} r & t \\ t & r \end{pmatrix} \begin{pmatrix} E_{\text{in},1} \\ E_{\text{in},2} \end{pmatrix} \tag{1}$$

where r is the self-coupling transmission coefficient (through port) and t is the cross-coupling coefficient (cross port).

3.1 1. The Real-Number Energy Contradiction

If both r and t are assumed to be real numbers, conservation of optical power requires S to be a **unitary matrix** ($S^\dagger S = I$, where $S^\dagger$ is the conjugate transpose):

$$S^\dagger S = \begin{pmatrix} r & t \\ t & r \end{pmatrix} \begin{pmatrix} r & t \\ t & r \end{pmatrix} = \begin{pmatrix} r^2 + t^2 & 2rt \\ 2rt & r^2 + t^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{2}$$

Setting off-diagonal terms to zero yields:

$$2rt = 0 \implies r = 0 \quad \text{or} \quad t = 0 \tag{3}$$

This implies either all light couples ($r = 0$) or no light couples ($t = 0$). Real-valued coefficients cannot represent a physical directional coupler without violating conservation of energy!

3.2 2. The Unitary Phase Fix

Assigning a 90° phase shift ($\pi/2$) to cross-coupled light by setting $t = jk$ (where k is real):

$$S = \begin{pmatrix} r & jk \\ jk & r \end{pmatrix} \implies S^\dagger = \begin{pmatrix} r & -jk \\ -jk & r \end{pmatrix} \tag{4}$$

Evaluating the unitary matrix product $S^\dagger S$:

$$S^\dagger S = \begin{pmatrix} r & -jk \\ -jk & r \end{pmatrix} \begin{pmatrix} r & jk \\ jk & r \end{pmatrix} = \begin{pmatrix} r^2 + (-jk)(jk) & r(jk) + (-jk)r \\ (-jk)r + r(jk) & (-jk)(jk) + r^2 \end{pmatrix} \tag{5}$$

Using $j^2 = -1$:

- **Off-Diagonal Terms:** $jrk - jrk = 0$ ✓
- **Diagonal Terms:** $r^2 - j^2k^2 = r^2 + k^2 = 1$ ✓

The matrix S is strictly unitary if and only if cross-coupling incorporates the quadrature phase factor j.

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4 Equation Walkthroughs & Derivations

4.1 Equation (1): Complex Electric Field Transmission

$$\frac{E_{\text{pass}}}{E_{\text{input}}} = e^{j(\pi+\phi)} \frac{a - re^{-j\phi}}{1 - rae^{j\phi}} \tag{6}$$

4.1.1 Plain Words

Equation (6) specifies the complex electric field transmission ratio at the output of the bus waveguide. It accounts for both transmission amplitude attenuation and phase delay imparted by recirculating ring interference.

4.1.2 Symbol Definitions

Symbol	Units	Physical Description
E_{pass}	V/m	Complex electric field amplitude exiting the bus waveguide
E_{input}	V/m	Complex electric field amplitude entering the bus waveguide
j	–	Imaginary unit ($\sqrt{-1}$)
ϕ	rad	Round-trip optical phase shift ($\phi = \beta L$)
β	rad/m	Modal propagation constant ($\beta = 2\pi n_{\text{eff}}/\lambda$)
L	m	Round-trip physical circumference ($L = 2\pi R$)
a	–	Single-pass amplitude transmission factor ($0 \leq a \leq 1$, $a = 1 \implies$ zero loss)
r	–	Self-coupling field amplitude coefficient ($0 \leq r \leq 1$)
k	–	Cross-coupling field amplitude coefficient ($r^2 + k^2 = 1$)

4.1.3 Assumptions

1. Monochromatic continuous-wave (CW), steady-state light propagation.
2. Single-mode optical propagation in both ring and bus waveguides.
3. Lossless directional coupler region ($r^2 + k^2 = 1$).
4. Negligible parasitic backscattering inside the ring or bus.

4.1.4 Step-by-Step Derivation

At the directional coupler junction, the input electric field E_{input} splits:

- Direct through field: rE_{input}
- Coupled field entering ring: jkE_{input}

Light in the ring completes a loop, accumulating single-pass attenuation factor a and phase shift ϕ, returning to the junction as $jkE_{\text{input}}ae^{j\phi}$. At the junction, a fraction r recirculates in the loop while a fraction jk cross-couples back into the bus.

Summing the direct pass and all subsequent recirculated optical leaks exiting into the bus:

$$\begin{aligned} E_{\text{pass}} &= rE_{\text{input}} + (jkE_{\text{input}})(ae^{j\phi})(jk) + (jkE_{\text{input}})(ae^{j\phi})(rae^{j\phi})(jk) + \dots \\ &= rE_{\text{input}} - k^2ae^{j\phi}E_{\text{input}}\left[1 + rae^{j\phi} + (rae^{j\phi})^2 + \dots\right] \end{aligned} \tag{7}$$

Summing the infinite geometric progression $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ (valid for $|rae^{j\phi}| < 1$):

$$\frac{E_{\text{pass}}}{E_{\text{input}}} = r - \frac{k^2ae^{j\phi}}{1 - rae^{j\phi}} = \frac{r(1 - rae^{j\phi}) - k^2ae^{j\phi}}{1 - rae^{j\phi}} = \frac{r - (r^2 + k^2)ae^{j\phi}}{1 - rae^{j\phi}} \tag{8}$$

Applying coupler power conservation ($r^2 + k^2 = 1$):

$$\frac{E_{\text{pass}}}{E_{\text{input}}} = \frac{r - ae^{j\phi}}{1 - rae^{j\phi}} \tag{9}$$

To convert this expression into standard textbook form, multiply numerator and denominator by $-e^{-j\phi}$:

$$\frac{E_{\text{pass}}}{E_{\text{input}}} = \frac{a - re^{-j\phi}}{1 - rae^{j\phi}} \cdot \left(-e^{j\phi}\right) \tag{10}$$

Recognizing that $-e^{j\phi} = e^{j\pi}e^{j\phi} = e^{j(\pi+\phi)}$, we arrive at **Equation** (6):

$$\frac{E_{\text{pass}}}{E_{\text{input}}} = e^{j(\pi+\phi)}\frac{a - re^{-j\phi}}{1 - rae^{j\phi}} \tag{11}$$

4.1.5 Numeric Sanity Check

Given representative silicon photonic design parameters:

- Ring Radius $R = 10\,\mu\text{m} \implies L = 2\pi R \approx 62.83\,\mu\text{m}$.
- Wavelength $\lambda = 1.55\,\mu\text{m}$, effective index $n_{\text{eff}} = 2.4 \implies \beta \approx 9.73 \times 10^6\,\text{rad/m}$.
- Waveguide Loss $\alpha = 2\,\text{dB/cm} \approx 46.05\,\text{m}^{-1} \implies a = e^{-\alpha L/2} \approx 0.9985$.
- Self-coupling coefficient $r = 0.95$.

At resonance ($\phi = 2m\pi \implies e^{j\phi} = 1$):

$$\frac{E_{\text{pass}}}{E_{\text{input}}} = e^{j\pi}\frac{0.9985 - 0.95}{1 - (0.95)(0.9985)} = -1 \cdot \frac{0.0485}{0.0514} \approx -0.9436 \tag{12}$$

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4.2 Equation (2): Optical Power Transmission Ratio (T_n)

$$T_n = \frac{I_{\text{pass}}}{I_{\text{input}}} = \frac{a^2 - 2ar\cos\phi + r^2}{1 - 2ar\cos\phi + (ra)^2} \tag{13}$$

4.2.1 Plain Words

Equation (13) defines the overall optical power transmission fraction T_n exiting the bus waveguide as a function of round-trip phase shift ϕ.

4.2.2 Step-by-Step Derivation

Optical power intensity is proportional to the squared magnitude of the electric field ($I \propto |E|^2$). Therefore, T_n is obtained by multiplying Eq. (6) by its complex conjugate:

$$T_n = \left| \frac{E_{\text{pass}}}{E_{\text{input}}} \right|^2 = \left(e^{j(\pi+\phi)} \frac{a - re^{-j\phi}}{1 - rae^{j\phi}} \right) \left(e^{-j(\pi+\phi)} \frac{a - re^{j\phi}}{1 - rae^{-j\phi}} \right) \tag{14}$$

Since $|e^{j(\pi+\phi)}|^2 = 1$, the phase term drops out:

$$T_n = \frac{(a - re^{-j\phi})(a - re^{j\phi})}{(1 - rae^{j\phi})(1 - rae^{-j\phi})} \tag{15}$$

Expanding numerator and denominator:
Numerator Expansion:

$$(a - re^{-j\phi})(a - re^{j\phi}) = a^2 - ar(e^{j\phi} + e^{-j\phi}) + r^2(e^{j\phi}e^{-j\phi}) \tag{16}$$

Using Euler's identity ($e^{j\phi} + e^{-j\phi} = 2\cos\phi$) and $e^{j\phi}e^{-j\phi} = 1$:

$$\text{Numerator} = a^2 - 2ar\cos\phi + r^2 \tag{17}$$

Denominator Expansion:

$$(1 - rae^{j\phi})(1 - rae^{-j\phi}) = 1 - ra(e^{j\phi} + e^{-j\phi}) + (ra)^2(e^{j\phi}e^{-j\phi}) \tag{18}$$

Applying Euler's identity:

$$\text{Denominator} = 1 - 2ar\cos\phi + (ra)^2 \tag{19}$$

Combining numerator and denominator yields **Equation** (13):

$$T_n = \frac{a^2 - 2ar\cos\phi + r^2}{1 - 2ar\cos\phi + (ra)^2} \tag{20}$$

4.2.3 Numeric Sanity Check

Using $a = 0.9985$ and $r = 0.95$:

- **On-Resonance ($\phi = 2m\pi \implies \cos\phi = 1$):**

$$T_n = \frac{(a-r)^2}{(1-ra)^2} = \frac{(0.9985 - 0.95)^2}{(1 - 0.95 \times 0.9985)^2} = \frac{(0.0485)^2}{(0.0514)^2} \approx 0.890 \quad (89.0\% \text{ power passed}) \tag{21}$$

- **Off-Resonance ($\phi = (2m+1)\pi \implies \cos\phi = -1$):**

$$T_n = \frac{(a+r)^2}{(1+ra)^2} = \frac{(0.9985 + 0.95)^2}{(1 + 0.95 \times 0.9985)^2} = \frac{(1.9485)^2}{(1.9486)^2} \approx 0.9999 \quad (\sim 100\% \text{ power passed}) \tag{22}$$

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4.3 Equation (7): Full Width at Half Maximum (FWHM)

$$\text{FWHM} = \frac{(1 - ra)\lambda_{\text{res}}^2}{\pi n_g L \sqrt{ra}} \tag{23}$$

4.3.1 Plain Words

This gives the Full Width at Half Maximum (FWHM) bandwidth of an all-pass ring resonance in units of wavelength. It tells you the physical spectral width of the transmission dip.

4.3.2 Symbol Definitions

Symbol	Units	Physical Description
FWHM	m (converted to nm)	Spectral width at half power depth
r	–	Self-coupling field coefficient
a	–	Single-pass amplitude transmission factor
λ_{res}	m	Resonant wavelength
n_g	–	Group index
L	m	Round-trip physical length

4.3.3 Assumptions

1. Narrow resonance approximation ($ra \approx 1$).
2. Includes first-order waveguide dispersion via n_g.

4.3.4 Derivation / Intuitive Justification

Setting the power transmission T_n in Eq. (13) equal to half its maximum dip depth allows solving for the phase width $\Delta\phi$. Converting $\Delta\phi$ to wavelength using $\Delta\lambda = \frac{\lambda^2}{\pi n_g L}\Delta\phi$ yields this expression.

4.3.5 Numeric Sanity Check

Given $\lambda_{\text{res}} = 1.55\,\mu\text{m}$, $L = 62.83\,\mu\text{m}$, $n_g = 4.2$, $r = 0.95$, $a = 0.9985 \implies ra \approx 0.9486$:

$$\text{FWHM} = \frac{(1 - 0.9486) \times (1.55 \times 10^{-6})^2}{\pi \times 4.2 \times (6.283 \times 10^{-5}) \times \sqrt{0.9486}} \approx \frac{0.0514 \times 2.4025 \times 10^{-12}}{8.073 \times 10^{-4}} \approx 1.53 \times 10^{-10}\,\text{m} = 0.153\,\text{nm} \tag{24}$$

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4.4 Equations (11) & (12): Extinction Transmission Bounds

$$T_t = \frac{(r + a)^2}{(1 + ra)^2} \quad \text{(Eq. 11)} \tag{25}$$

$$R_{\text{min}} = \frac{(r - a)^2}{(1 - ra)^2} \quad \text{(Eq. 12)} \tag{26}$$

4.4.1 Plain Words

Eq. (25) calculates the maximum off-resonance power transmission (T_t), and Eq. (26) calculates the minimum on-resonance power transmission (R_{min}) for an all-pass ring filter. The ratio T_t / R_{min} gives the On-Off Extinction Ratio (ER).

4.4.2 Symbol Definitions

Symbol	Units	Physical Description
T_t	–	Maximum off-resonance power transmission
$R_{\min}$	–	Minimum on-resonance power transmission
r	–	Self-coupling coefficient
a	–	Round-trip loss amplitude factor

4.4.3 Assumptions

1. All-pass ring resonator configuration.
2. Continuous-wave (CW) operation.

4.4.4 Derivation / Intuitive Justification

Derived directly from Eq. (13) by setting $\cos\phi = -1$ for maximum off-resonance transmission (T_t) and $\cos\phi = 1$ for minimum on-resonance transmission ($R_{\min}$).

4.4.5 Numeric Sanity Check

For $r = 0.95$ and $a = 0.9985$:

$$T_t = \frac{(0.95 + 0.9985)^2}{(1 + 0.95 \times 0.9985)^2} \approx 0.9999 \tag{27}$$

$$R_{\min} = \frac{(0.95 - 0.9985)^2}{(1 - 0.95 \times 0.9985)^2} = \frac{(-0.0485)^2}{(0.0514)^2} \approx 0.8895 \tag{28}$$

$$\text{Extinction Ratio} = 10\log_{10}\left(\frac{T_t}{R_{\min}}\right) = 10\log_{10}\left(\frac{0.9999}{0.8895}\right) \approx 0.51\,\text{dB} \tag{29}$$

If critically coupled ($r = a = 0.95$), $R_{\min} = 0 \implies$ Extinction Ratio $= \infty$ dB.

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5 Design Synthesis & Critical Coupling

Key Physical Takeaway

For an all-pass microring filter, Eq. (13) demonstrates that transmitted power drops to **absolute zero** ($T_n = 0$) at resonance if and only if:

$$a = r \tag{30}$$

This condition is known as **Critical Coupling**. At critical coupling, internal single-pass optical attenuation (a) perfectly equals self-coupling transmission (r). The optical field leaking out of the ring matches the amplitude of the uncoupled bus field but is 180° out of phase, producing complete destructive interference.

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Coupling Regime	Mathematical Condition	On-Resonance Behavior
Under-Coupled	$r > a$	Partial dip in transmission ($T_n > 0$). Ring loss exceeds coupling leakage.
Critically Coupled	$r = a$	Complete extinction ($T_n = 0$). Perfect destructive interference.
Over-Coupled	$r < a$	Partial dip in transmission ($T_n > 0$). High coupling leakage, broader resonance linewidth.

6 Check Your Understanding

Check Your Understanding

Question 1: If a microring resonator is completely lossless ($a = 1$) with self-coupling $r = 0.9$, what is the power transmission T_n on resonance ($\phi = 2m\pi$)?

Answer: Substituting $a = 1$ into Eq. (13):

$$T_n = \frac{(1 - 0.9)^2}{(1 - 0.9)^2} = 1.0 \quad (100\% \text{ transmission}) \tag{31}$$

Explanation: Without internal absorption or scattering loss, light circulating in the ring eventually leaks entirely back into the forward bus waveguide direction, leaving net power unchanged.

Question 2: What occurs to the on-resonance power transmission T_n when the coupling gap is mechanically or thermally adjusted such that $r = a$?

Answer: When $r = a$, the numerator $(a - r)^2 = 0$, giving $T_n = 0$ ($-\infty$ dB extinction ratio). This represents the critical coupling condition.